

# Data science

## *The Big Data challenge*

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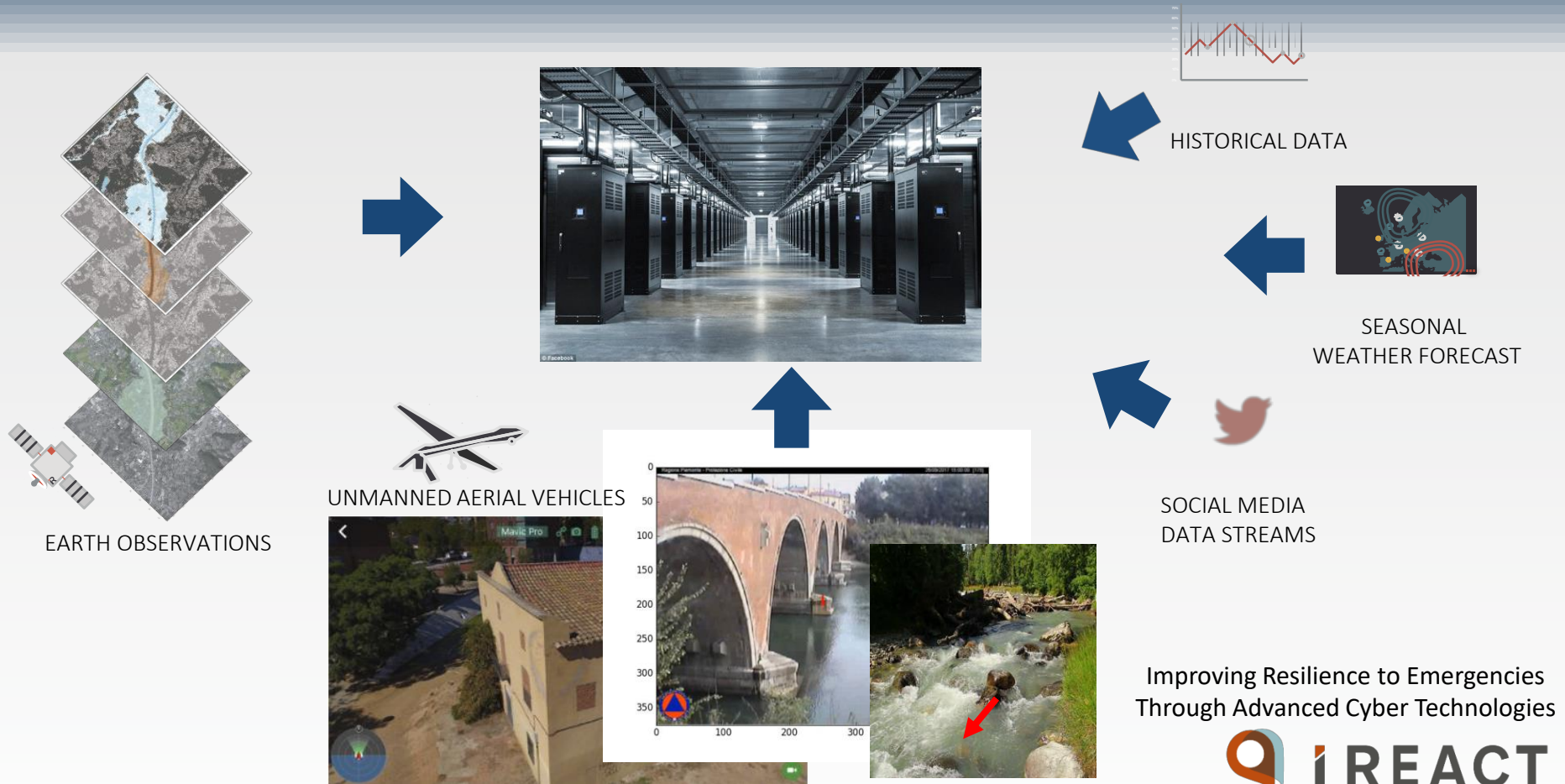


Data Base and Data Mining Group of Politecnico di Torino

# Big data hype?



# Emergency management



Improving Resilience to Emergencies  
Through Advanced Cyber Technologies

**Q i REACT**

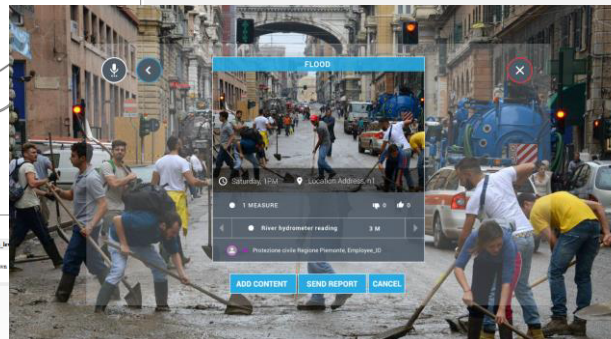
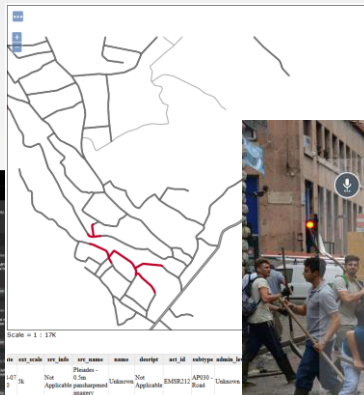
# Emergency management



FIRST RESPONDERS AND  
DECISION MAKERS



CITIZENS



Improving Resilience to Emergencies  
Through Advanced Cyber Technologies





# User engagement

2005



2013



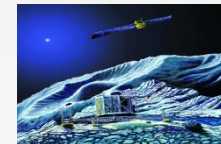
# Who generates big data?

## ❑ User Generated Content (Web & Mobile)

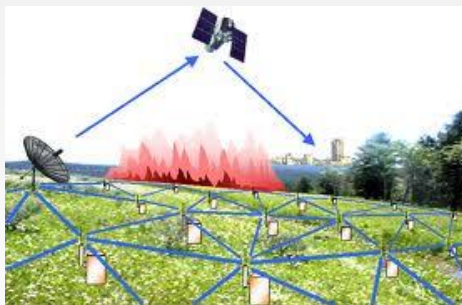
❑ E.g., Facebook, Instagram, Yelp, TripAdvisor, Twitter, YouTube



## ❑ Health and scientific computing



- ❑ Log files
  - ❑ Web server log files, machine syslog files
- ❑ Internet Of Things
  - ❑ Sensor networks, RFID, smart meters



# What is big data?



## □ Many different definitions

*“Data whose scale, diversity and complexity require new architectures, techniques, algorithms and analytics to manage it and extract value and hidden knowledge from it”*



# What is big data?



□ Many different definitions

*“Data whose **scale**, **diversity** and **complexity** require new architectures, techniques, algorithms and analytics to manage it and extract value and hidden knowledge from it”*

# What is big data? Huge data collection



□ Many different definitions

*“Data whose **scale**, **diversity** and **complexity** require new **architectures**, **techniques**, **algorithms** and **analytics** to manage it and extract value and hidden knowledge from it”*

# What is big data?



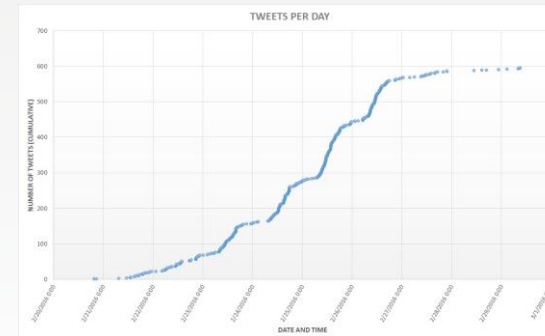
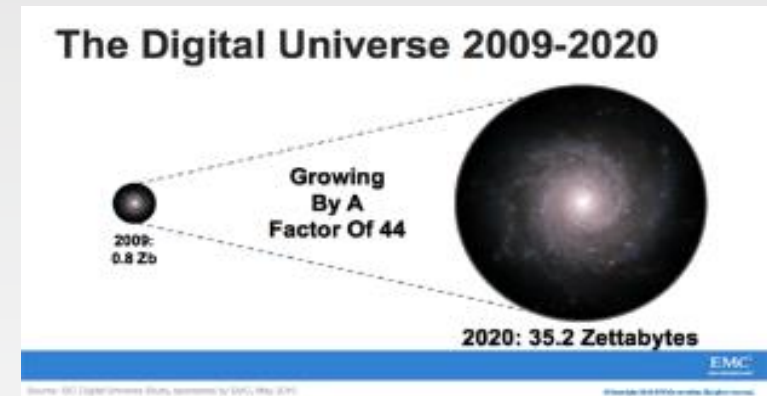
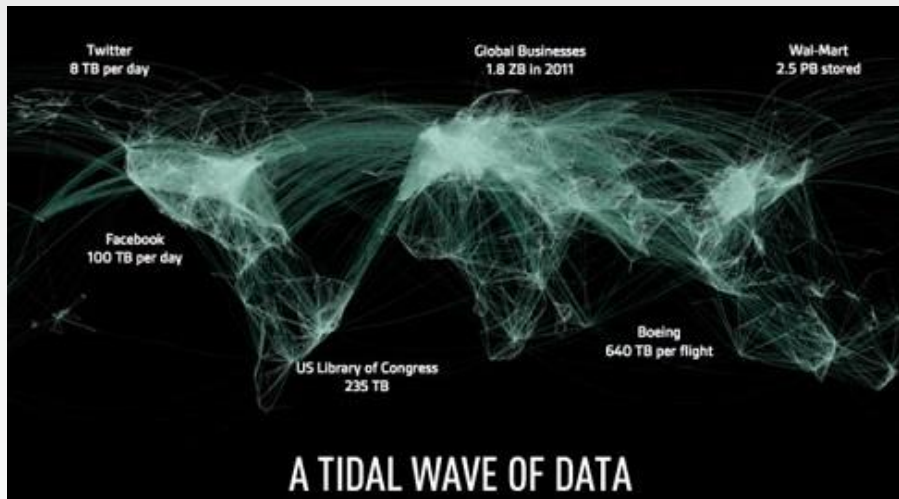
□ Many different definitions

*“Data whose scale, diversity and complexity require new architectures, techniques, algorithms and analytics to manage it and extract **value** and hidden **knowledge** from it”*



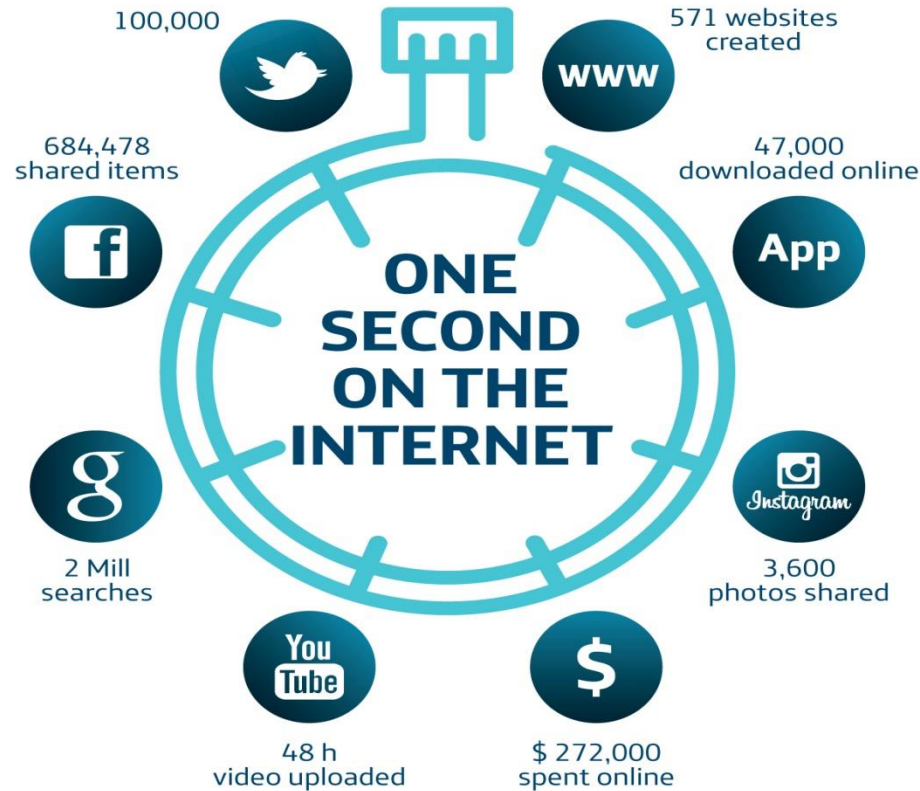
# The Vs of big data: **V**olume

- ❑ Data volume increases exponentially over time
- ❑ 44x increase from 2009 to 2020
- ❑ Digital data 35 ZB in 2020





# On the Internet...



Source: Telefónica analysis based on Social and Digital Media Revolution Statistics 2013 from MistMediaGroup (<http://youtube.com/watch?v=Slb5x5fixk4>).

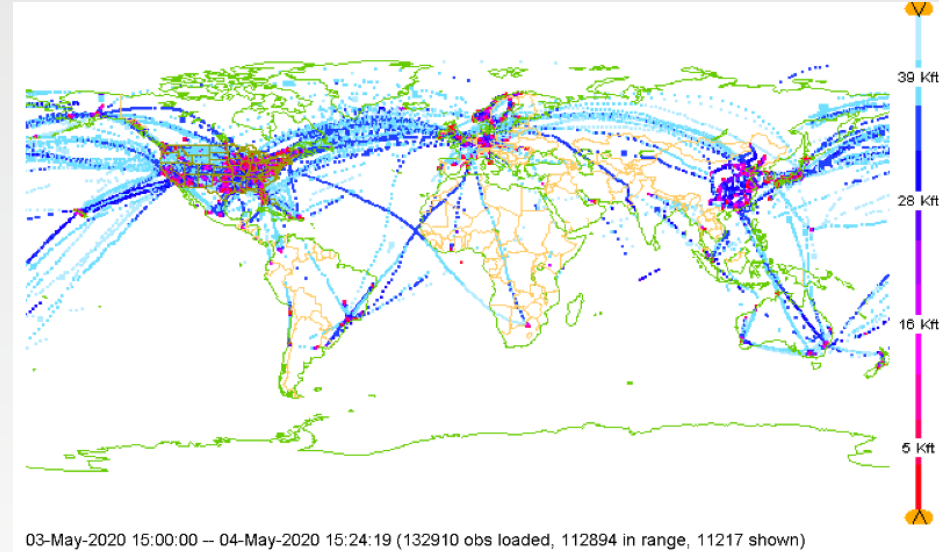
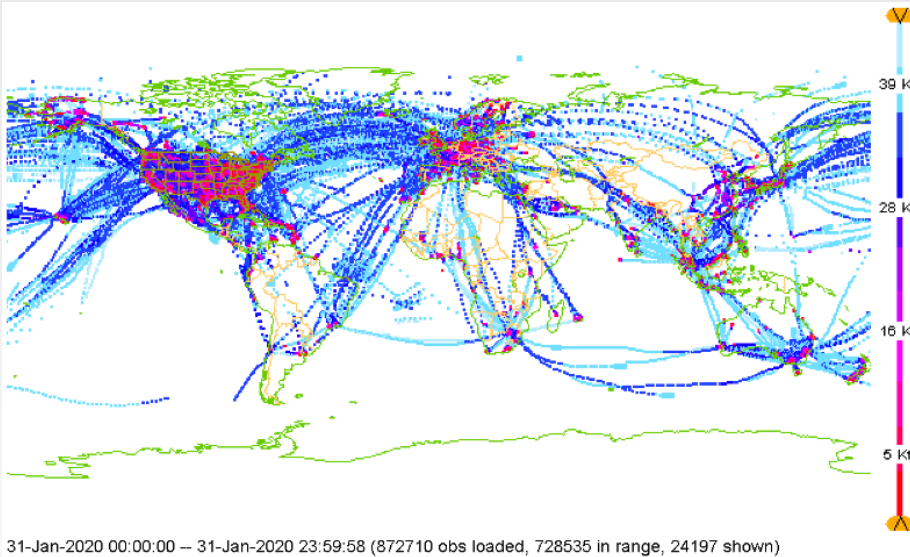
- <http://www.internetlivestats.com/>

# Weather forecast



January 2020

May 2020



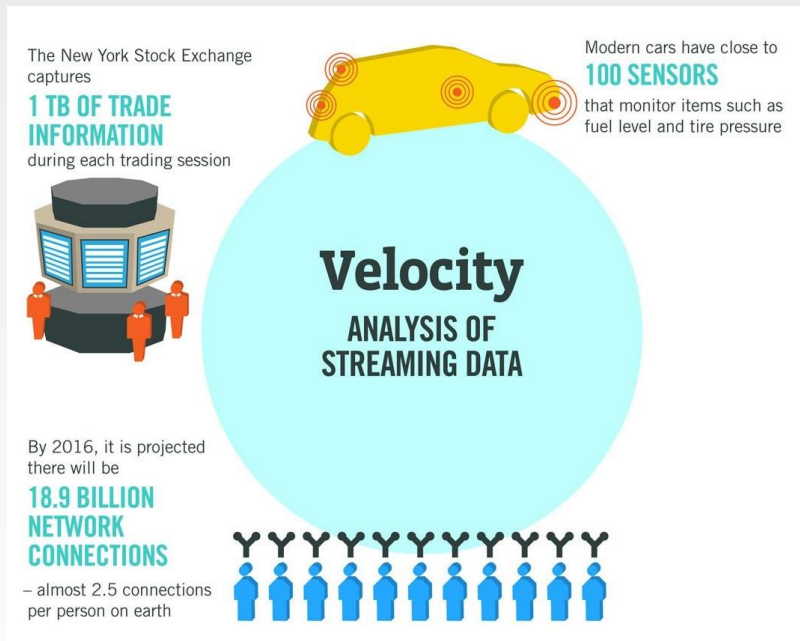
# The Vs of big data: Velocity

- Fast data generation rate

  - Streaming data

- Very fast data processing to ensure timeliness

We continuously collect data in a very high frequency. Usually, data need to be analyzed by real-time processes.





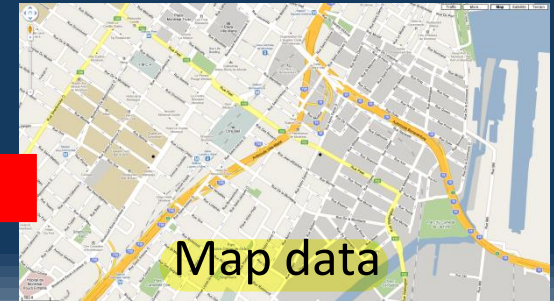
# (Near) Real time processing



Crowdsourcing



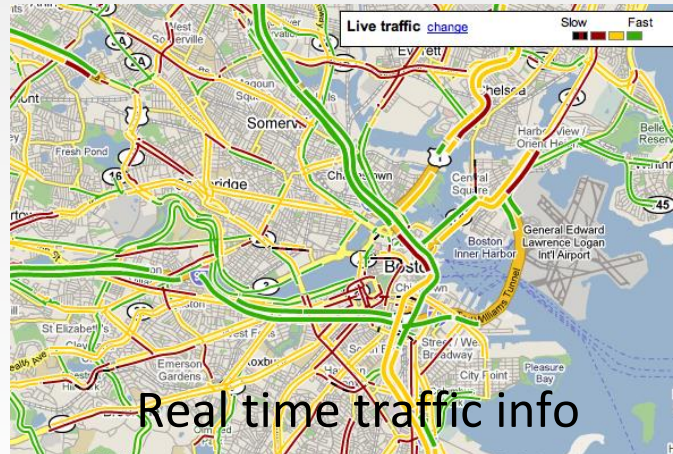
Computing



Map data



Sensing



Real time traffic info

● Data collection  
● Data processing

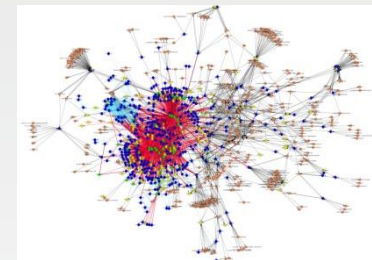
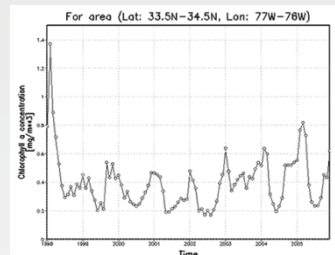
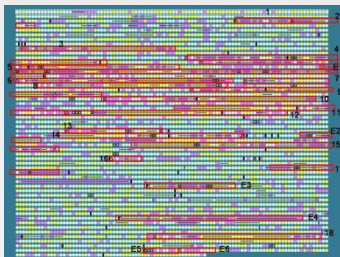


# The Vs of big data: Variety

- ❑ Various formats, types and structures

- ❑ Numerical data, image data, audio, video, text, time series

*We have to consider heterogeneous data collection that need to be properly processed.*



- ❑ A single application may generate many different formats

# The Vs of big data: **Veracity**

□ Data quality

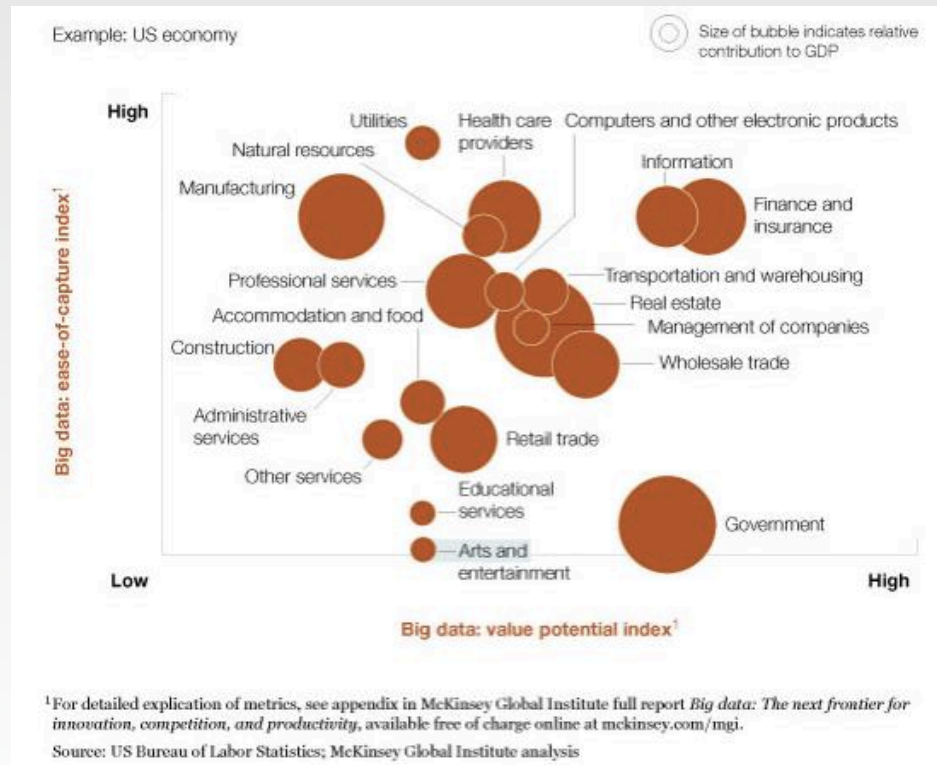


We must be able to process data which contains high quality values.

# The most important V: Value

□ Translate data into business advantage

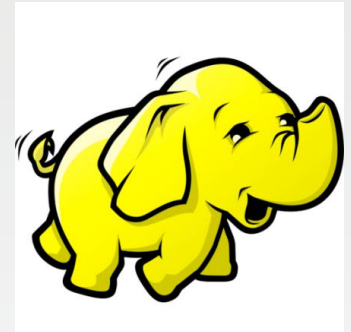
The entire process has a sense only if at the end we obtain a value.



# Big data challenges

- ❑ Technology & infrastructure
  - ❑ New architectures, programming paradigms and techniques

*Transfer the processing power to the data*
  - ❑ Apache Hadoop/Spark ecosystem
- ❑ Data management & analysis
  - ❑ New emphasis on “data”

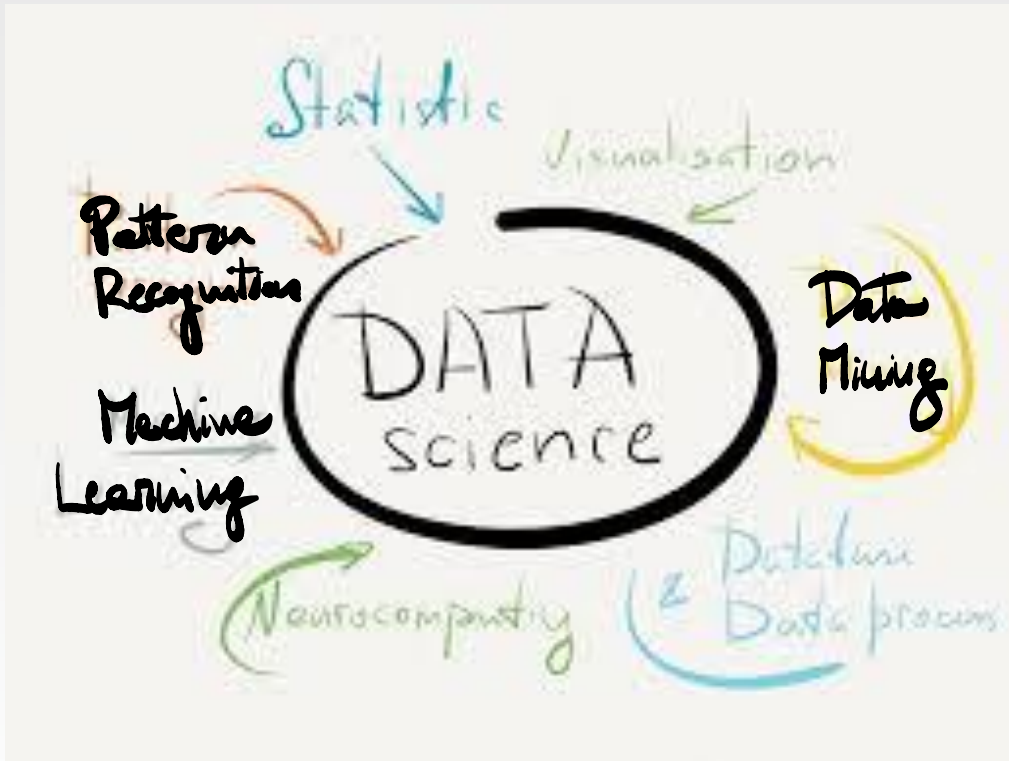


➔ ***Data science***



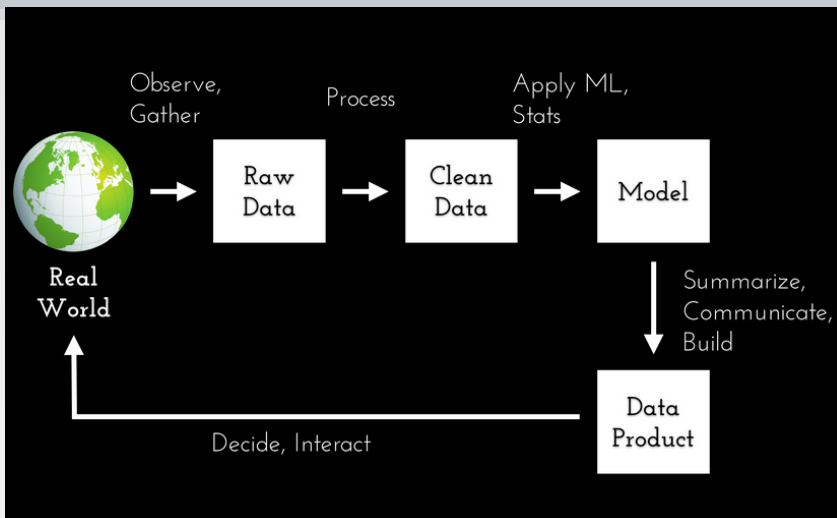
# Data science

“Extracting meaning from very large quantities of data”



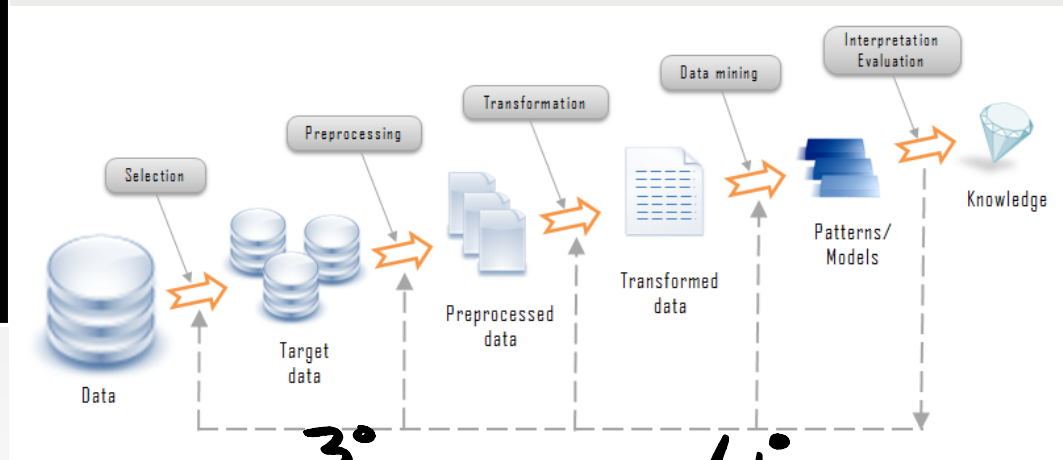
D.J. Patil coined the word *data scientist*

# The data science process



AKA **KDD** process

**K**nowledge **D**iscovery in **D**atabases



Steps:

1°

2°

3°

4°

Generation

Acquisition

Storage

Analysis

# Generation

- ❑ Passive recording
  - ❑ Typically structured data
  - ❑ Bank trading transactions, shopping records, government sector archives
- ❑ Active generation
  - ❑ Semistructured or unstructured data
  - ❑ User-generated content, e.g., social networks
- ❑ Automatic production
  - ❑ Location-aware, context-dependent, highly mobile data
  - ❑ Sensor-based Internet-enabled devices (IoT)



# Acquisition

## ☐ Collection

- ☐ Pull-based, e.g., web crawler
- ☐ Push-based, e.g., video surveillance, click stream

## ☐ Transmission

- ☐ Transfer to data center over high capacity links

## ☐ Preprocessing

- ☐ Integration, cleaning, redundancy elimination





# Storage

- ❑ Storage infrastructure
  - ❑ Storage technology, e.g., HDD, SSD
  - ❑ Networking architecture, e.g., DAS, NAS, SAN
- ❑ Data management
  - ❑ File systems (HDFS), key-value stores (Memcached), column-oriented databases (Cassandra), document databases (MongoDB)
- ❑ Programming models
  - ❑ Map reduce, stream processing, graph processing



# Analysis

## ❑ Objectives

- ❑ Descriptive analytics, predictive analytics, prescriptive analytics

## ❑ Methods

- ❑ Statistical analysis, machine learning and data mining, text mining, network and graph data mining
- ❑ Association analysis, classification and regression, clustering
- ❑ Diverse domains call for customized techniques



# Machine learning and data mining

- ❑ Non trivial extraction of

- ❑ implicit
  - ❑ previously unknown
  - ❑ potentially useful

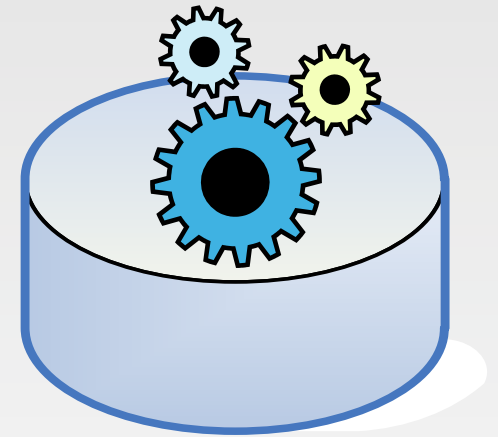
information from available data

- ❑ Extraction is automatic

- ❑ performed by appropriate algorithms

- ❑ Extracted information is represented by means of abstract models

- ❑ denoted as *pattern*



# Example: profiling

- ❑ Consumer behavior in e-commerce sites

- ❑ Selected products, requested information, ...



- ❑ Search engines and portals

- ❑ Query keywords, searched topics and objects



- ❑ Social network data

- ❑ Profiles (Facebook, Instagram, ...)

- ❑ Dynamic data: posts on blogs, FB, tweets



- ❑ Maps and georeferenced data

- ❑ Localization, interesting locations for users



# Example: profiling

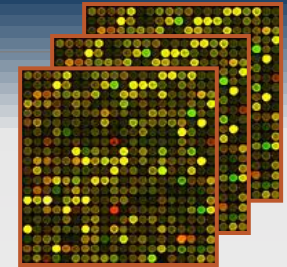
- ❑ User/service profiling
  - ❑ Recommendation systems, advertisements
- ❑ Market basket analysis
  - ❑ Correlated objects for cross selling
  - ❑ User registration, fidelity cards
- ❑ Context-aware data analysis
  - ❑ Integration of different dimensions
  - ❑ E.g., location, time of the day, user interest
- ❑ Text mining
  - ❑ Brand reputation, sentiment analysis, topic trends



# Example: biological data

## Microarray

- expression level of genes in a cellular tissue
- various types (mRNA, DNA)



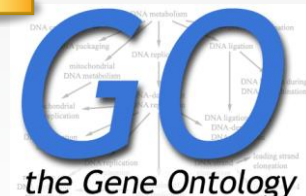
## Patient clinical records

- personal and demographic data
- exam results

| CLID     | PATIENT ID  | shx013: 49A34 | shx060: 45A9 | shx077: 52A28 | shx009: 4A34 | shx014: 61A31 | shx082: 99A6 | shx083: 46A15 | shx008: 41A31 |
|----------|-------------|---------------|--------------|---------------|--------------|---------------|--------------|---------------|---------------|
| IMAGE:74 | ISG20    in | -1.02         | -2.34        | 1.44          | 0.57         | -0.13         | 0.12         | 0.34          | -0.51         |
| IMAGE:76 | TNFSF13     | -0.52         | -4.06        | -0.29         | 0.71         | 1.03          | -0.67        | 0.22          | -0.09         |
| IMAGE:36 | LOC93343    | -0.25         | -4.08        | 0.06          | 0.13         | 0.08          | 0.06         | -0.08         | -0.05         |
| IMAGE:23 | ITGA4    in | -1.375        | -1.605       | 0.155         | -0.015       | 0.035         | -0.035       | 0.505         | -0.865        |

## Textual data in public collections

- heterogeneous formats, different objectives
- scientific literature (PubMed)
- ontologies (Gene Ontology)



# Biological analysis objectives

## □ Clinical analysis

- detecting the causes of a pathology
- monitoring the effect of a therapy

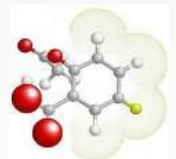
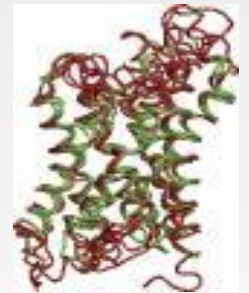
⇒ diagnosis improvement and definition of new specific therapies

## □ Bio-discovery

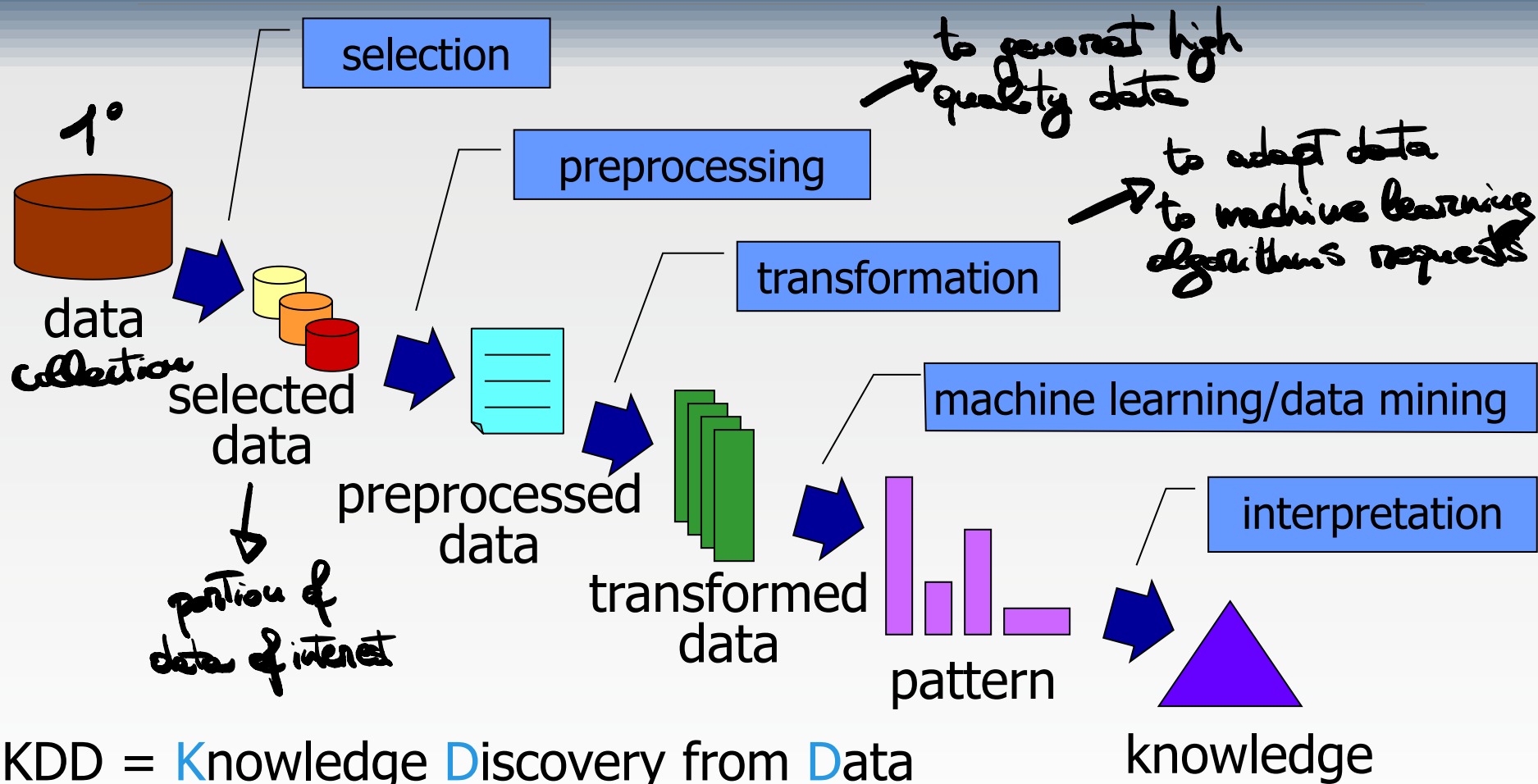
- gene network discovery
- analysis of multifactorial genetic pathologies

## □ Pharmacogenesis

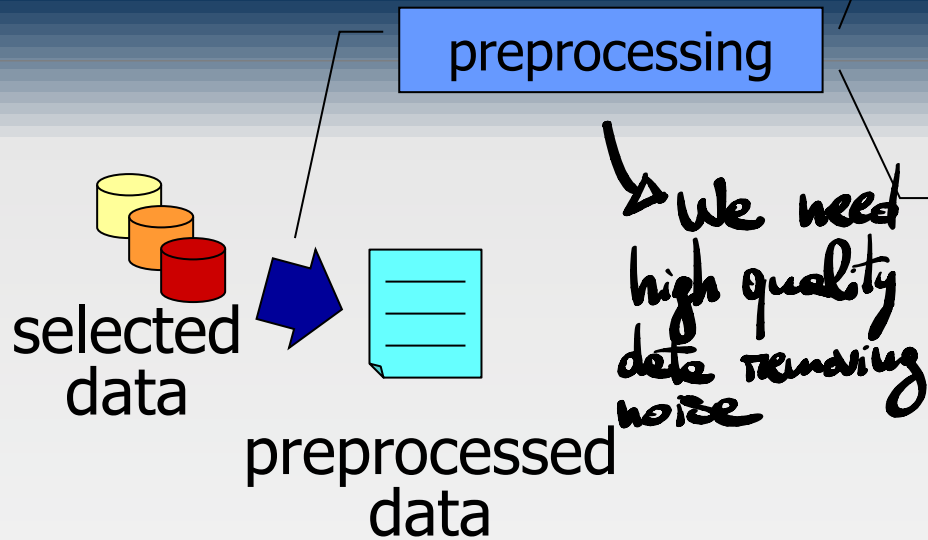
- lab design of new drugs for genic therapies



# Knowledge Discovery Process



# Preprocessing



## data cleaning

- reduces the effect of noise
- identifies or removes outliers
- solves inconsistencies

*season errors*

## data integration

- reconciles data extracted from different sources
- integrates metadata
- identifies and solves data value conflicts
- manages redundancy

*out  
scale  
values*

Real world data is "dirty"

Without good quality data, no good quality pattern

# A word from practitioners

- ❑ At least 80-90% of their work involves not machine learning, but
  - ❑ Working with experts to understand the domain, assumptions, questions
  - ❑ Trying to catalog and make sense of the data sources
  - ❑ Wrangling, extracting, and integrating the data
  - ❑ Cleaning the wrangled data



# Association rules

## ❑ Objective

- ❑ extraction of frequent correlations or pattern from a transactional database



Tickets at a supermarket counter

| TID | Items                      |
|-----|----------------------------|
| 1   | Bread, Coke, Milk          |
| 2   | Beer, Bread                |
| 3   | Beer, Coke, Diapers, Milk  |
| 4   | Beer, Bread, Diapers, Milk |
| 5   | Coke, Diapers, Milk        |
| ... | ...                        |

## ■ Association rule

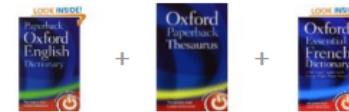
diapers  $\Rightarrow$  beer

- 2% of transactions contains both items
- 30% of transactions containing diapers also contain beer

# Association rules



## Frequently Bought Together



Price For All Three: £9.00

Add all three to Basket

Show availability and delivery details

- ☒ **This item:** Paperback Oxford English Dictionary by Oxford Dictionaries Paperback £3.00
- ☒ Oxford Paperback Thesaurus by Oxford Dictionaries Paperback £3.00
- ☒ Oxford Essential French Dictionary by Oxford Dictionaries Paperback £3.00

## Jobs You May Be Interested In

Powered by  
LinkedIn



**Senior Data Analyst Job**  
Thomson Reuters - Bangalore, KA



**Data Scientist/ Senior Data Scientist**  
HeadHonchos.com - Bangalore - IN

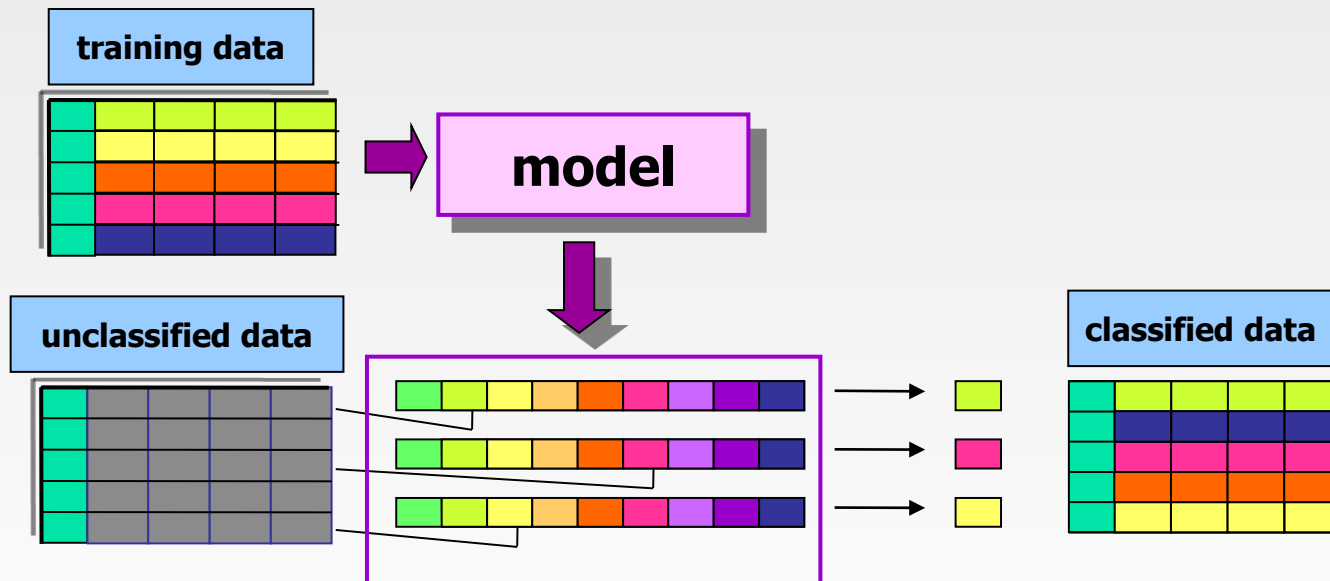


**Hiring Computer Scientist (Java) for...**  
Adobe - Noida

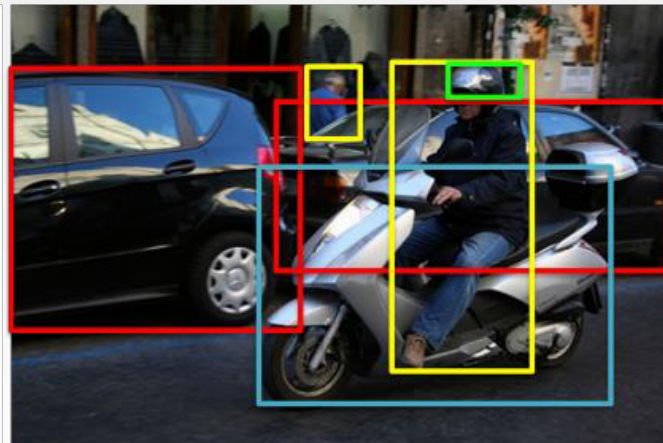
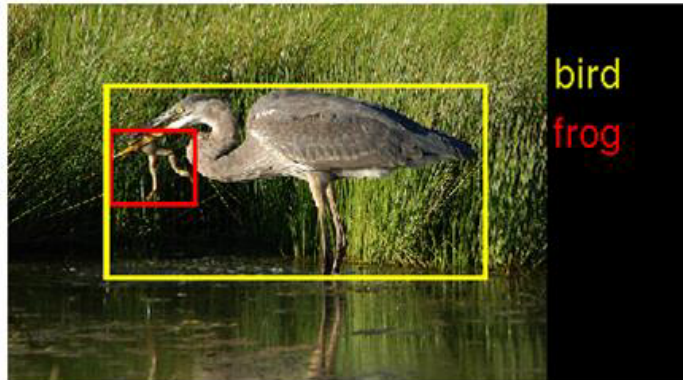
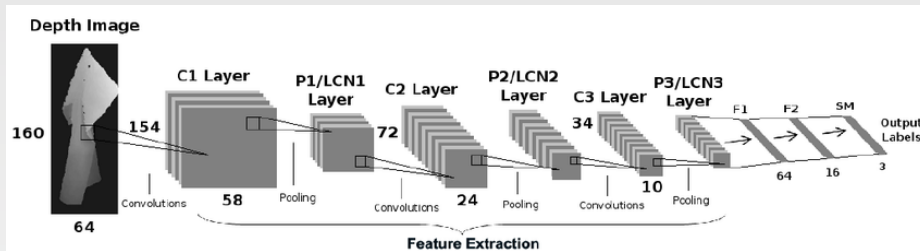
# Classification

## Objectives

- prediction of a class label
- definition of an interpretable model of a given phenomenon



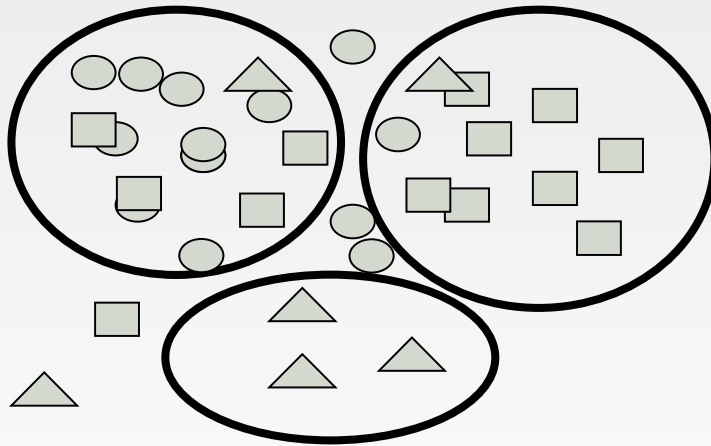
# Classification



Person  
Car  
Motorcycle  
Helmet

# Clustering

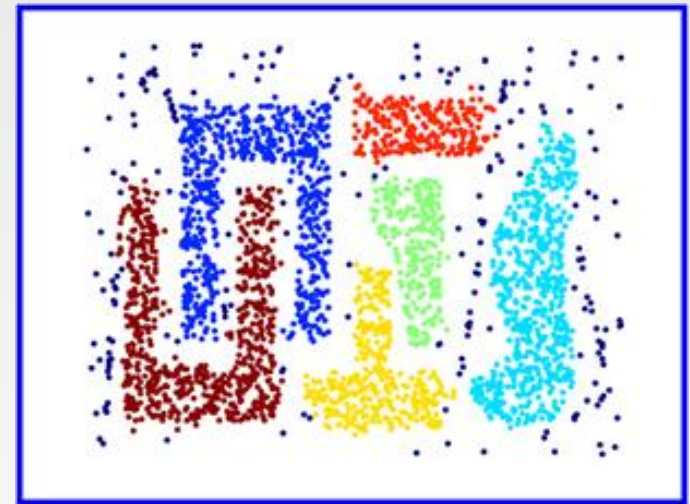
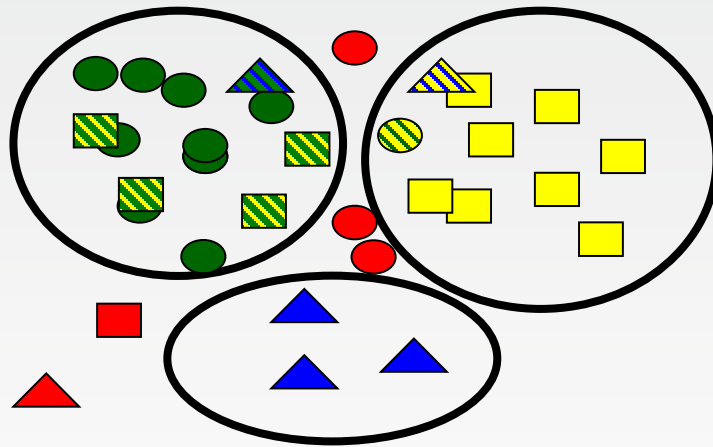
- ❑ Objectives
  - ❑ detecting groups of similar data objects
  - ❑ identifying exceptions and outliers



# Clustering

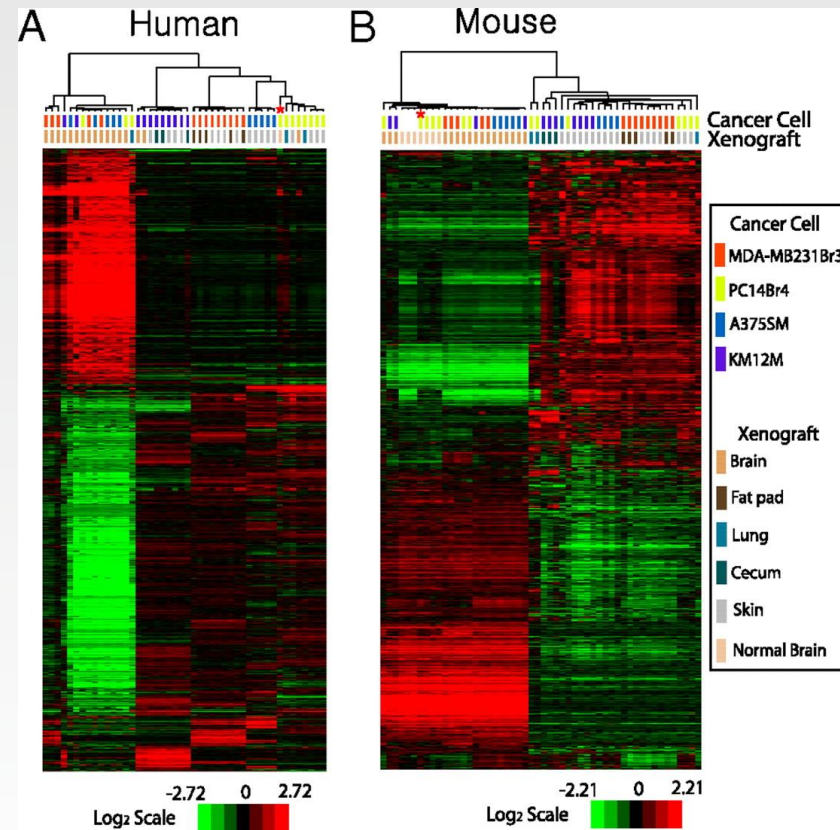
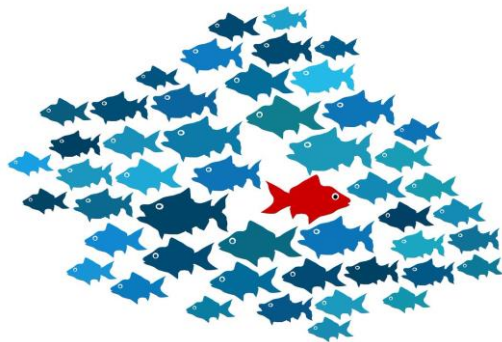
## Objectives

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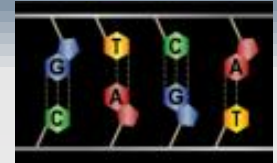
# Clustering



# Other data mining techniques

## ❑ Sequence mining

- ❑ ordering criteria on analyzed data are taken into account
- ❑ example: motif detection in proteins



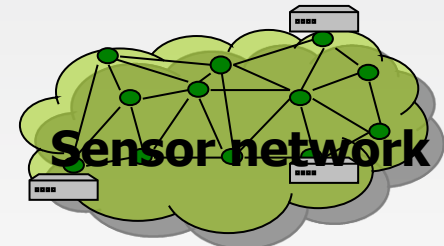
## ❑ Time series and geospatial data

- ❑ temporal and spatial information are considered
- ❑ example: sensor network data



## ❑ Regression

- ❑ prediction of a continuous value
- ❑ example: prediction of stock quotes



## ❑ Outlier detection

- ❑ example: intrusion detection in network traffic analysis



# The data science process

- ☐ What *question* are you answering?
- ☐ What is the right *scope* of the project?
- ☐ What *data* will you use?
- ☐ What *techniques* are you going to try?
- ☐ How will you *evaluate* your result?
- ☐ What *maintenance* will be required?

# The data science recipe

## ❑ Different ingredients needed

### ❑ Data expert

- ❑ Data processing, data structures

### ❑ Data analyst

- ❑ Data mining, statistics, machine learning

### ❑ Visualization expert

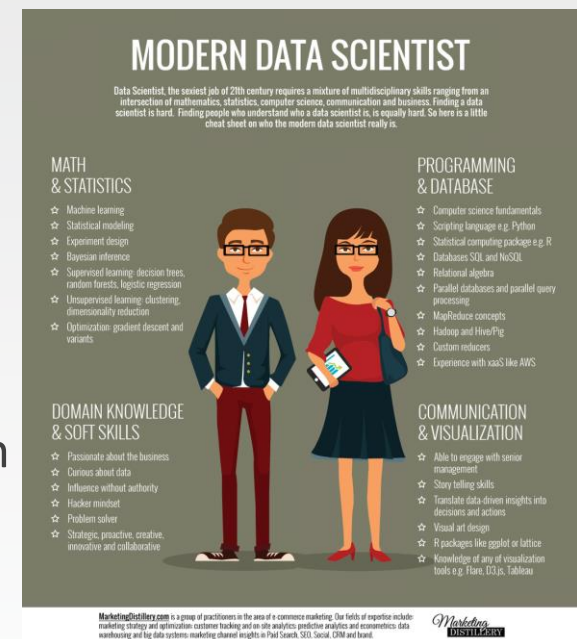
- ❑ Visual art design, storytelling skills

### ❑ Domain expert

- ❑ Provide understanding of the application domain

### ❑ Business expert

- ❑ Data driven decisions, new business models



# Open issues

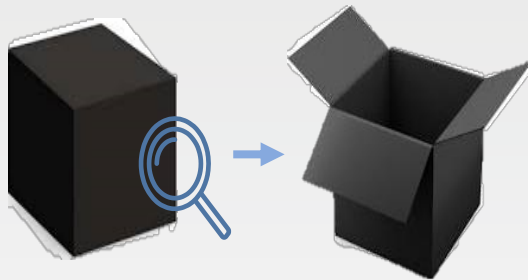
- ❑ Social impact of analysis is very important
  - ❑ Interpretability and transparency of the analysis process
  - ❑ Bias in algorithms and data
  - ❑ Privacy preservation





# Interpretability in machine learning

*“The ability to explain or to present in understandable terms to a human”*



Open the black box



Trade-off Accuracy-Interpretability

- ❑ **Model explanation:** global understanding of how a model works
- ❑ **Prediction explanation:** local understanding of why a prediction is made
- ❑ **Interpretable feature selection:** incorporating interpretability-based criteria into the model design



# Interpretability

- ❑ Learned decision rule in pneumonia patients dataset from USA hospital

*history of asthma → lower chance of dying from pneumonia*

- ❑ MD consider asthma as a serious risk factor for people who get pneumonia

- ❑ Analysis

- ❑ asthmatics probably notice earlier the symptoms of pneumonia
- ❑ a healthcare professional is going to provide earlier pneumonia diagnosis
- ❑ as high-risk patients, they're going to get high-quality treatment sooner than other people

➡ asthmatics actually have almost half the chance of dying of non-asthmatics

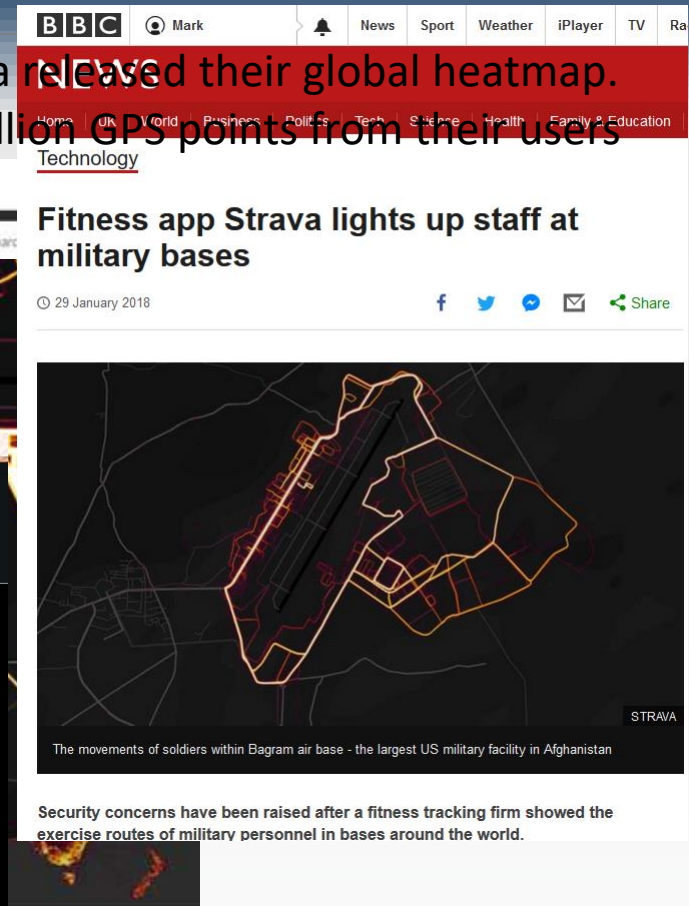
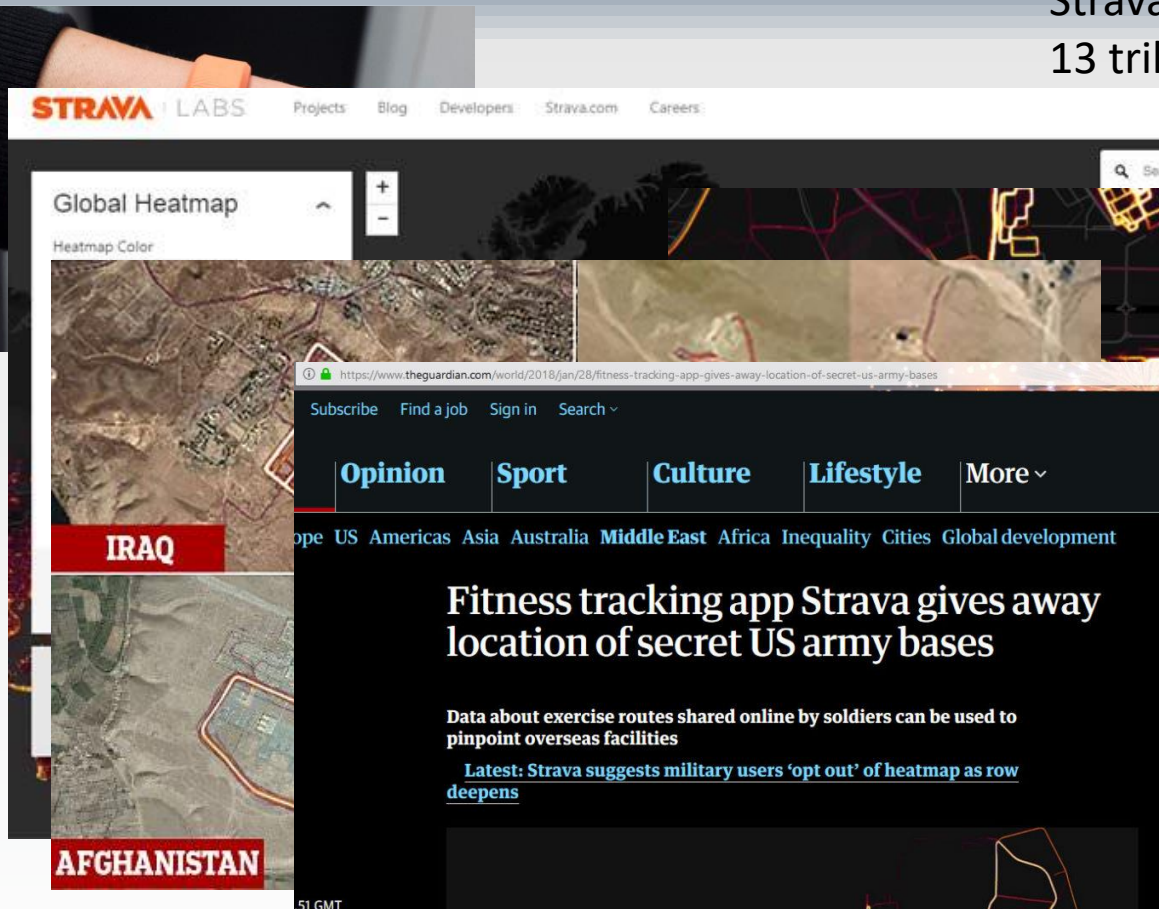
- ❑ Using a neural network, this model issue would *never* have been uncovered

# Algorithmic and data bias

- ❑ Task: predict likelihood of an individual committing a future crime
  - ❑ Risk scores used by US criminal justice system
- ❑ Scores computed from
  - ❑ Questions answered by the defendants
  - ❑ Information pulled by criminal records
- ❑ Race was not among the questions
  - ❑ ... however other items may be correlated (e.g., poverty, joblessness)
- ❑ Software product flagged black defendants as future criminals more frequently than white defendants
  - ➡ Training data was biased by a larger black defendant population

# Privacy

Strava released their global heatmap.  
13 trillion GPS points from their users



# Open issues

- ❑ Social impact of analysis is very important
  - ❑ Interpretability and transparency of the analysis process
  - ❑ Privacy preservation
- ❑ Many technical issues are not solved
  - ❑ Scalability to *huge* data volumes
  - ❑ Data dimensionality
  - ❑ Complex data structures, heterogeneous data formats
  - ❑ Data quality
  - ❑ Streaming data